

KINETICS & REACTOR DESIGN

CHE F311 · BITS Pilani · I-Semester 2026-27

Short Notes & Formula Sheet

Mole balances · Sizing · Rate laws · Isothermal design
Rate-data analysis · Multiple reactions · Non-isothermal design
Catalysis · Non-ideal reactors and residence-time distributions

Nine decks, 163 slides

Compiled from: the nine course slide decks in Slides/ — Ch1-krd (1).pdf (12 pp), Ch2-krd (1).pdf (15 pp), Ch3-krd.pdf (16 pp), Ch4-Krd.pdf (23 pp), Ch5-_krd.pdf (19 pp), Ch6_krd.pdf (18 pp), Ch7-_krd.pdf (24 pp), Catalysis_Krd.pdf (14 pp) and Non_ideal_KRD.pdf (22 pp).

Reference text: H. S. Fogler, *Elements of Chemical Reaction Engineering*; O. Levenspiel, *Chemical Reaction Engineering* (the decks' own problem sources).

Companion documents: KRD-Problems-and-Solutions.pdf works every numerical printed in these decks; krd-answers-check.py reproduces every number quoted below and exits 0.

One caveat about a source: Non_ideal_KRD.pdf is headed *Reaction Engineering / CHE G641* — the graduate version of this material. It is the only deck in the set that does not carry the CHE F311 heading, and section 9 says where that shows up.

What is inside

Topic	Deck
Mole balances, rate of reaction, conversion, the four design equations	Ch1
Reactor sizing from conversion; the Levenspiel plot; numerical integration; reactors in series and in parallel; optimum size ratio	Ch2
Rate laws and stoichiometry; relative rates; units of k ; elementary vs non-elementary and reversible laws; Arrhenius; stoichiometric tables for batch and flow, liquid and gas phase	Ch3
Isothermal reactor design; the algorithm; closed-form conversions; Damköhler number; pressure drop in packed beds	Ch4
Collection and analysis of rate data; differential and integral methods; half-life; differential reactor; nonlinear regression	Ch5
Multiple reactions; selectivity and yield; contacting schemes for the four sign cases; series intermediates; product distribution; net rates	Ch6
Non-isothermal design; energy balance; adiabatic operation; equilibrium thermodynamics; heat exchange; multiple steady states and runaway; interstage cooling and optimum feed temperature	Ch7
Catalysis and catalytic reactors; adsorption and site balance; Langmuir–Hinshelwood; internal and external diffusion	Catalysis
Non-ideal reactors; residence-time distribution; segregation; tanks-in-series; dispersion; two-parameter models	Non-ideal

Notation used throughout

X	conversion of A	τ	space time, V/v_0
$-r_A$	rate of disappearance of A	Da	Damköhler number
F_{A0}, F_A	molar flow of A in / out	k, A, E	rate constant, pre-exponential, activation energy
C_A, C_{A0}	concentration of A / at inlet	ε, δ	volume-change factor, mole-change number
V, W	reactor volume, catalyst mass	K_C, X_e	equilibrium constant, equilibrium conversion
N_A, N_{A0}	moles of A / initially	ΔH_{Rx}	heat of reaction
Θ_i	C_{i0}/C_{A0} , feed ratio	Ua, T_a	heat-transfer coefficient×area, coolant temperature
p	P/P_0 , reduced pressure	ϕ, η	Thiele modulus, effectiveness factor
α	pressure-drop parameter	$E(t), F(t)$	RTD function, cumulative RTD
t_m, σ^2	mean residence time, variance	Pe, n	Péclet number, tanks-in-series number
$S_{D/U}, \tilde{S}_{D/U}$	selectivity, instantaneous / overall	$\tilde{Y}_D$	yield, rate basis / mole basis

1 Mole Balances, Rates of Reaction and Conversion

Source: Ch1-krd (1).pdf.

1.1 Rate of reaction and the bases it can be written on

- **Rate of reaction** is the number of moles of a species reacting per unit time per unit *extensive quantity* of the system. The extensive quantity is what distinguishes the bases:

$-r_A = \text{mol}/(\text{volume} \cdot \text{time})$	homogeneous: per reactor volume V
$-r'_A = \text{mol}/(\text{mass cat.} \cdot \text{time})$	heterogeneous: per catalyst mass W
$r''_A = \text{mol}/(\text{area} \cdot \text{time})$	per catalyst surface area S
$-r'_A W = r''_A S = -r_A V \quad \implies \quad \boxed{\text{all three give the same molar rate}}$	

Choosing the wrong basis is the most common error in packed-bed work: the design equation must be written on the same basis as the rate law.

1.2 Conversion, extent and the general mole balance

$X = \frac{\text{moles of A reacted}}{\text{moles of A fed}}$	conversion, defined on the limiting reactant
$N_i = N_i^0 + \nu_i \alpha$	extent of reaction α
$F_{j0} - F_j + G_j = \frac{dN_j}{dt}$	general mole balance on species j

- G_j is the rate of generation of j by reaction within the system volume — $G_j = r_j V$. Everything else in the balance is a transport term.
- Conversion is always defined on the **limiting** reactant, so that $0 \leq X \leq 1$.

1.3 The four design equations

Batch: $\frac{dN_A}{dt} = r_A V$	$\implies t = N_{A0} \int_0^X \frac{dX}{-r_A V}$
CSTR: $V = \frac{F_{A0} X}{(-r_A)_{\text{exit}}}$	rate evaluated at <i>exit</i> conditions
PFR: $\frac{dF_A}{dV} = r_A$	$\implies V = F_{A0} \int_0^X \frac{dX}{-r_A}$
PBR: $\frac{dF_A}{dW} = r'_A$	$\implies W = F_{A0} \int_0^X \frac{dX}{-r'_A}$

- The structural difference between a **CSTR** and a **PFR** is exactly the difference between evaluating $-r_A$ at the exit and integrating it along the path. That single fact drives all of section 2.
- **Space time** $\tau = V/v_0$; **space velocity** $= v_0/V = 1/\tau$.

1.4 Worked from the deck

- **Rocket engine** (slide 36). Chamber 75×60 cm, exhaust 108 kg/s.

$V = 0.2121 \text{ m}^3$	$\implies$	$-r_{\text{H}_2} = 28 \text{ kmol m}^{-3} \text{ s}^{-1}, \quad -r_{\text{O}_2} = 14 \text{ kmol m}^{-3} \text{ s}^{-1}$
--------------------------	------------	--

The 2:1 ratio of the two rates is the stoichiometry of $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ — the first appearance of the relative-rate rule developed in section 3.

- **A→B at three duties** (slide 54). The deck asks for the volume of a CSTR and a PFR, and the batch time, at successively harsher conditions:

	(a)	(b)	(c)
CSTR	99.0 dm ³	2750 dm ³	66 000 dm ³
PFR	99.0 dm ³	127.9 dm ³	660 dm ³
Batch time	9.90 h	12.79 h	66.0 h

In (a) the two reactors are *equal* because the rate is constant there; from (b) onward the CSTR grows far faster than the PFR, which is the Levenspiel-plot argument of the next section in numerical form.

2 Conversion and Reactor Sizing

Source: Ch2-krd (1).pdf.

2.1 The Levenspiel plot

- Plot $F_{A0}/(-r_A)$ **against** X . Then the two design equations become areas on that single graph:

$$V_{\text{CSTR}} = \text{rectangle} = \left. \frac{F_{A0}}{-r_A} \right|_X \cdot X \quad \text{height at the exit} \times \text{width}$$

$$V_{\text{PFR}} = \text{area under the curve} = \int_0^X \frac{F_{A0}}{-r_A} dX \quad \text{integral from 0 to } X$$

- For any *positive-order* kinetics $-r_A$ falls as X rises, so $F_{A0}/(-r_A)$ rises and the **rectangle always exceeds the area under the curve**: a CSTR needs more volume than a PFR for the same conversion. The deck's own numbers in §1.4 show the gap opening from 0 to a factor of 100.

2.2 Numerical integration of the PFR integral

$$\begin{aligned} \text{Simpson (3 pt):} \quad \int &\approx \frac{h}{3} (f_0 + 4f_1 + f_2) \\ \text{Simpson (5 pt):} \quad \int &\approx \frac{h}{3} (f_0 + 4f_1 + 2f_2 + 4f_3 + f_4) \\ \text{3/8 rule:} \quad \int &\approx \frac{3h}{8} (f_0 + 3f_1 + 3f_2 + f_3) \end{aligned}$$

The five-point Simpson rule is the one the deck uses throughout, and it is the one that reproduces the printed answers.

2.3 Reactors in series

$$\begin{aligned} F_{Ai} &= F_{A0} (1 - X_i) && \text{flow leaving stage } i \\ V_i &= \frac{F_{A0} (X_i - X_{i-1})}{(-r_A)_i} && \text{CSTR } i : \text{rectangle from } X_{i-1} \text{ to } X_i \\ V_i &= F_{A0} \int_{X_{i-1}}^{X_i} \frac{dX}{-r_A} && \text{PFR } i : \text{area over the same interval} \end{aligned}$$

2.4 Reactors in parallel

- Rule:** for the total volume to be a minimum the feed must be split so that V/F (**equivalently** τ) is the **same in every branch**. Branches running at different τ waste volume.

2.5 Choosing the best arrangement

- Work from the $1/(-r_A)$ **versus** C_A curve.
- $n > 1$ (rate falls steeply): plug flow first, then a small mixed reactor, then a large mixed reactor.
- $n < 1$: the order is reversed — mixed flow first.
- Two CSTRs, minimum total volume:** maximise the inscribed rectangle, which gives the tangency condition $-dy/dx = y/x$ on the $1/(-r_A)$ curve.

2.6 Worked from the deck — the sizing table (slides 19–38)

The deck gives $T = 500$ K, $P = 830$ kPa, $F_{A0} = 0.4$ mol/s and this rate table. Note that the third row is $-r_A$ itself, not its reciprocal — the reciprocal is the last column.

X	0	0.1	0.2	0.4	0.6	0.7	0.8
$-r_A$ (mol m ⁻³ s ⁻¹)	0.45	0.37	0.30	0.195	0.113	0.079	0.05
$F_{A0}/(-r_A)$ (m ³)	0.89	1.08	1.33	2.05	3.54	5.06	8.00

$$V_{\text{CSTR}} = 8.00 \times 0.8 = 6.40 \text{ m}^3 \quad \text{rectangle to } X = 0.8$$

$$V_{\text{PFR}} = \frac{0.2}{3} (0.89 + 4(1.33) + 2(2.05) + 4(3.54) + 8.00) = \boxed{2.165 \text{ m}^3} \quad \text{5-pt Simpson}$$

$$V_{\text{CSTR}} - V_{\text{PFR}} = 6.40 - 2.165 = 4.235 \text{ m}^3 \quad \text{the price of back-mixing}$$

- **Two CSTRs in series** (slide 9), split at $X = 0.4$:

$$V_1 = 2.05 \times 0.4 = 0.82 \text{ m}^3, \quad V_2 = 8.00 \times (0.8 - 0.4) = 3.20 \text{ m}^3 \Rightarrow \boxed{4.02 \text{ m}^3}$$

- **Two PFRs in series** (slide 10), same split:

$$V_1 = 0.551 \text{ m}^3, \quad V_2 = 1.614 \text{ m}^3 \Rightarrow \boxed{2.165 \text{ m}^3} \quad (\text{identical to one PFR})$$

Two points worth keeping. First, splitting a PFR changes nothing — the integral is additive — whereas splitting a CSTR cuts the volume from 6.40 to 4.02 m³, a 37% saving. Second, the series-CSTR total still exceeds the PFR, and keeps falling as more stages are added, approaching the PFR area as $N \rightarrow \infty$.

- **Unknown stoichiometry from an MFR** (slide 42). $C_{A0} = 0.10$, $C_{B0} = 0.01$ mol/L, $V = 1$ L, outlet 0.02/0.03/0.04:

$$r_A = -0.08, \quad r_B = +0.02, \quad r_C = +0.04 \Rightarrow \boxed{4A \rightarrow B + 2C}$$

- **Three PFRs in two parallel branches** (slide 46). Branch D holds 50 + 30 L, branch E holds 40 L; equal τ requires

$$\text{fraction to D} = \frac{50}{90} = \boxed{0.556}$$

- **P2-7** (slide 59), $1/(-r_A) = 1.0, 1.67, 5.0, 5.0, 5.0, 5.0, 1.25, 0.91$ at $X = 0, 0.2, 0.4, 0.45, 0.5, 0.6, 0.8, 0.9$:

- (a) CSTR 24.0 dm³; PFR 71.9 dm³ for $X = 0.8$
- (b) the two volumes are equal at $X \approx 0.69$
- (c) a 105 dm³ CSTR gives $X = 0.70$
- (d) CSTR then PFR: $X_1 = 0.401$, $X_2 = 0.643$
- (e) PFR then CSTR: $X_1 = 0.400$, $X_2 = 0.900$

(d) and (e) are the same two reactors in the opposite order and differ by 0.26 in exit conversion — the $1/(-r_A)$ curve here is *non-monotonic*, so the ordering rule of §2.5 actually bites.

3 Rate Laws and Stoichiometry

Source: Ch3-krd.pdf.

3.1 Relative rates of reaction



$$\frac{r_A}{-a} = \frac{r_B}{-b} = \frac{r_C}{c} = \frac{r_D}{d} \Rightarrow \boxed{r_B = \frac{b}{a} r_A}$$

Divide each species' rate by its own stoichiometric coefficient *with sign*; the results are all equal. This is the check that closes the rocket-engine problem of §1.4.

3.2 Form of the rate law

$-r_A = k C_A^\alpha C_B^\beta$	power law; overall order $n = \alpha + \beta$
$k = A e^{-E/RT}$	Arrhenius
$\ln \frac{k_2}{k_1} = \frac{E}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$	two-temperature form
$K_C = \frac{k_f}{k_r}$	elementary reversible; $-r_A \rightarrow 0$ at equilibrium

- **α and β are empirical** — they are *not* the stoichiometric coefficients unless the reaction is elementary. Determining them is the whole subject of section 5.
- **Units of k (slide 9):** $k = (\text{concentration})^{1-n} / \text{time}$, so $\text{mol dm}^{-3} \text{s}^{-1}$ (zero), s^{-1} (first), $\text{dm}^3 \text{mol}^{-1} \text{s}^{-1}$ (second), $\text{dm}^6 \text{mol}^{-2} \text{s}^{-1}$ (third) — the units alone betray the order.
- **Arrhenius plot (slides 27–28):** $\ln k = \ln A - (E/R)(1/T)$ is a straight line in $1/T$; a larger E means a steeper slope, i.e. a reaction that is far more temperature-sensitive — high- E reactions surge with a small temperature rise yet crawl at low temperature.

3.3 Elementary, non-elementary and reversible rate laws

- **Elementary (slide 14):** the orders *are* the stoichiometric coefficients, as in $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ with $-r_{\text{NO}} = k C_{\text{NO}}^2 C_{\text{O}_2}$; the deck notes there is no known reaction with molecularity 4. Some non-elementary reactions nonetheless *follow* an elementary rate law (the orders happen to match the coefficients).
- **Non-elementary (slides 15–17):** homogeneous $\text{CO} + \text{Cl}_2 \rightarrow \text{COCl}_2$ with $r_{\text{CO}} = k C_{\text{CO}} C_{\text{Cl}_2}^{3/2}$ (orders 1, $\frac{3}{2}$, overall $\frac{5}{2}$); heterogeneous toluene hydrogenation $-r'_T = k P_T P_H / (1 + K_B P_B + K_T P_T)^2$ and cumene hydrogenolysis $-r'_C = k P_C P_H / (1 + K_B P_B + K_C P_C)$, with K_B, K_T adsorption constants; enzymatic sucrose $\rightarrow$ glucose + fructose (Michaelis–Menten form).
- **Reversible (slides 18–21):** the rate law must reduce to the thermodynamic relation $K_C = C_C^c C_D^d / (C_A^a C_B^b)$ at equilibrium. The deck's benzene dimerisation, $2\text{C}_6\text{H}_6 \rightleftharpoons \text{C}_{12}\text{H}_{10} + \text{H}_2$: forward $k_B C_B^2$, reverse $k_{-B} C_D C_H$, net

$$-r_B = k_B \left(C_B^2 - \frac{C_D C_H}{K_C} \right), \quad \frac{k_B}{k_{-B}} = K_C.$$

3.4 Stoichiometric tables

For $aA + bB \rightarrow cC + dD$, with $\Theta_i = C_{i0}/C_{A0}$.

Batch	$N_A = N_{A0}(1 - X), \quad N_B = N_{A0}(\Theta_B - \frac{b}{a}X), \quad N_C = N_{A0}(\Theta_C + \frac{c}{a}X)$
Batch, liquid	$V = V_0$, so $C_i = N_i/V_0$
Flow	the same expressions with $N \rightarrow F$
Batch, gas, const. P, T	$V = V_0(1 + \varepsilon X), \quad C_A = \frac{N_{A0}(1 - X)}{V_0(1 + \varepsilon X)}$

$$C_i = C_{A0} \frac{\Theta_i + \nu_i X/a}{1 + \varepsilon X} \cdot \frac{T_0}{T} \cdot \frac{P}{P_0} \quad \text{gas-phase flow, general}$$

$$\varepsilon = y_{A0} \delta, \quad \delta = \frac{c + d - b - a}{a} \quad \text{mole change on complete conversion}$$

- $\varepsilon = 0$ whenever there is no change in mole number, or the feed is diluted enough that y_{A0} is small. Setting $\varepsilon = 0$ is the single most common legitimate simplification, and the deck's gas-phase examples all state it explicitly.
- $\varepsilon > 0$ (moles increase) means the gas expands as conversion proceeds, concentrations fall faster than stoichiometry alone implies, and the reactor needs more volume than the constant-density calculation suggests.

3.5 Worked from the deck

- **Relative rates** (slides 5–7).

$$\begin{aligned} r_{\text{NO}_2} = +4 &\implies r_{\text{NO}} = -4, & r_{\text{O}_2} = -2 \text{ mol m}^{-3} \text{s}^{-1} \\ r_A = -10 &\implies r_B = -15, & r_C = +25 \text{ mol dm}^{-3} \text{s}^{-1} \end{aligned}$$

- **Dimerisation in an MFR** (slide 61). Four runs give

$$X = 0.2502, 0.4996, 0.6667, 0.7995; \quad -r_A = 2502.2, 1498.9, 800.0, 399.8 \text{ mol L}^{-1}\text{h}^{-1}$$

$$\boxed{n = 1.96 \approx 2}, \quad k = 0.3407, 0.3369, 0.3200, 0.3583 \text{ L mmol}^{-1}\text{h}^{-1}$$

The scatter in k across the four runs — 0.320 to 0.358, about $\pm 6\%$ — is the practical test that the assumed order is right: a wrong order makes k drift systematically instead of scattering.

- **Phosphine decomposition** (slide 63). $4\text{PH}_3 \rightarrow \text{P}_4 + 6\text{H}_2$, first order, $k = 10/\text{h}$, 649°C , 460 kPa , 40 mol/h , 80% conversion:

$$C_{A0} = 60.00 \text{ mol m}^{-3} \quad \Rightarrow \quad \boxed{V = 0.1882 \text{ m}^3 \approx 188 \text{ L}}$$

C_{A0} comes from P/RT at the stated temperature and pressure — the deck supplies the state, not the concentration.

- **Milk sterilisation** (slide 61). The same result in 30 min at one temperature and 15 s at the other means the same k ; with first-order kinetics,

$$E = \frac{R \ln(t_1/t_2)}{1/T_1 - 1/T_2} = \frac{8.314 \ln(1800/15)}{1/347.15 - 1/353.15} \Rightarrow \boxed{E \approx 813 \text{ kJ/mol}}$$

The deck prints “ 80°C for 30 min” beside “ $74^\circ\text{C} \dots 15 \text{ s}$ ”; a hotter treatment needs the *shorter* time, so the printed pairing is reversed and is reported as such — the calculation pairs 353.15 K with 15 s and 347.15 K with 1800 s . The enormous E is exactly why pasteurisation is so sensitive to temperature.

4 Isothermal Reactor Design

Source: Ch4-Krd.pdf.

4.1 The algorithm

Five steps, in this order, every time.

1. **Mole balance** — pick the reactor, write its design equation (section 1).
2. **Rate law** — from the data or the mechanism (section 3).
3. **Stoichiometry** — express every C_i in terms of X (the tables of §3.4).
4. **Combine** — substitute 2 and 3 into 1 to get one equation in X .
5. **Evaluate** — integrate, or solve, and size the reactor.

Every worked example in this section follows those five steps; the deck labels them explicitly on the slides.

4.2 Closed-form conversions

Batch, 1st order	$t = \frac{1}{k} \ln \frac{1}{1-X}$
Batch, 2nd order	$t = \frac{1}{kC_{A0}} \cdot \frac{X}{1-X}$
CSTR, 1st order	$X = \frac{k\tau}{1+k\tau}$
CSTR, 2nd order	$X = \frac{(1+2Da) - \sqrt{1+4Da}}{2Da}, \quad Da = k\tau C_{A0}$

$$Da_1 = k\tau$$

$$Da \leq 0.1 \Rightarrow X < 0.1$$

$$Da \geq 10 \Rightarrow X > 0.9$$

Damköhler number, first order

reaction-limited: little conversion

near-complete conversion

Da is the ratio of the reaction rate to the convective transport rate. It is the quickest way to see, before any integration, whether a proposed reactor is remotely big enough.

4.3 Reactors in series and in parallel

$$N \text{ equal CSTRs, 1st order: } X = 1 - \frac{1}{(1+k\tau_i)^N}$$

Parallel CSTRs: behave as one large CSTR of the same total volume

Compare with §2.6: there, splitting the duty into two CSTRs saved 37% of the volume. This is the same result in closed form, and it is why industrial trains are staged rather than single-vessel.

4.4 Variable volume, gas phase

$$V_{\text{PFR}} = \frac{v_0}{kC_{A0}} \left[2\varepsilon(1+\varepsilon)\ln(1-X) + \varepsilon^2 X + \frac{(1+\varepsilon)^2 X}{1-X} \right] \quad \text{2nd order, gas phase}$$

4.5 Pressure drop in packed beds

$$\begin{aligned} \frac{dp}{dW} &= -\frac{\alpha}{2p}, & p &= \frac{P}{P_0} & \text{reduced-pressure form} \\ \varepsilon = 0 : \quad p &= (1 - \alpha W)^{1/2} & & \text{closed form} \\ C_A &= C_{A0} \frac{1-X}{1+\varepsilon X} p & & \text{gas, constant } T \\ \frac{dP}{dz} &= -\frac{\beta_0 P_0}{P} (1 + \varepsilon X), & \beta_0 &= \frac{G(1-\phi)}{\rho_0 D_p \phi^3} \left[\frac{150(1-\phi)\mu}{D_p} + 1.75 G \right] \end{aligned}$$

- ϕ is the bed porosity, G the superficial mass flux, D_p the particle diameter. β_0 is the **Ergun equation** in the form the deck uses.
- Because C_A carries a factor p , pressure drop *slows the reaction*, so the conversion achieved in a given catalyst mass is lower than the no-drop calculation predicts. The two must be solved **coupled**, not sequentially.

4.6 Worked from the deck

- **Ethylene oxide, batch** (slides 17–20). Regression of the concentration–time data gives

$$k = 0.315 \text{ min}^{-1}, \quad t_{90\%} = 7.32 \text{ min}$$

and, at the much slower $k = 10^{-4} \text{ s}^{-1}$,

$$t_{90\%} = \boxed{6.40 \text{ h}}$$

- **Ethylene glycol, CSTR** (slides 29–38):

$$F_{A0} = 7.672 \text{ lbmol/min}, \quad v_0 = 15.34 \text{ ft}^3/\text{min}, \quad \boxed{V_{80\%} = 197.3 \text{ ft}^3}$$

Arrangement	τ (min)	Da	X
two CSTRs in parallel	13.94	4.34	0.813
two CSTRs in series	6.97	2.168	0.900

Same total volume, and the series arrangement reaches 0.900 against 0.813 — the staging argument of §4.3 with the deck's own numbers.

- **Ethylene cracking, PFR** (slide 44). 300 M lb/yr, 1100 K, 6 atm, $k = 0.072 \text{ s}^{-1}$ at 1000 K, $E = 82 \text{ kcal/mol}$:

$$k(1100 \text{ K}) = 3.067 \text{ s}^{-1}, \quad F_{A0} = 154.1 \text{ mol/s}, \quad C_{A0} = 66.476 \text{ mol/m}^3$$

$$v_0 = 2.318 \text{ m}^3/\text{s} \quad \Rightarrow \quad \boxed{V = 1.83 \text{ m}^3 = 64.6 \text{ ft}^3}$$

3150 tubes of 0.0205 ft^3 each. The Arrhenius step is what moves k from 0.072 to 3.067 s^{-1} — a factor of 43 for 100 K.

- **Packed bed, $A + B \rightarrow 2C$** (slides 55–58), $\alpha = 0.0099$, $W = 100$:

$$\boxed{X = 0.602 \text{ with pressure drop}} \quad \text{vs.} \quad \boxed{X = 0.75 \text{ without}}$$

Ignoring pressure drop overstates conversion by 0.15, a quarter of the achieved value. This is the number to quote when justifying a coupled solution.

- **PBR and CSTR in either order** (slide 76):

Arrangement	group	X
PBR then CSTR	0.02471	0.598
CSTR then PBR	0.02471	0.690
PBR then CSTR, $\alpha' = 0.00711$	0.02471	0.727

Exit $p = 0.3162$, i.e. $P = 6.32$ atm. Putting the CSTR *first* gains 0.09 in conversion, because the CSTR then operates at full pressure and the PBR only sees the already-reduced rate. Reducing α' recovers most of the loss — pressure drop, not ordering, is the dominant effect.

- **Series reactions in batch**, $A \rightarrow B \rightarrow C$ (slide 78):

$$t_{\max} = 0.693 \text{ h}, \quad C_{B,\max} = 1.250$$

- **Pressure-drop length check** (slides 90–98). Without pressure drop, 16.8 m reaches $X = 0.10$ and 35.5 m reaches $X = 0.20$.

A discrepancy in the source, reported rather than smoothed over. The deck states that about 20 m achieves the target *with* pressure drop, and asks for $X = 0.20$ in a 1.5 cm bed. The no-drop calculation already needs 35.5 m for $X = 0.20$, and adding pressure drop can only make the bed longer — so $X = 0.20$ is not reachable in 1.5 cm. The deck's own figures are reproduced above and the conflict is flagged here rather than resolved by adjusting a number.

5 Collection and Analysis of Rate Data

Source: Ch5-_krd.pdf.

5.1 Differential method

- Take logs of the power law to linearise it:

$$\ln\left(-\frac{dC_A}{dt}\right) = \ln k + \alpha \ln C_A \quad \text{slope} = \alpha, \text{ intercept} = \ln k$$

The derivatives come from the concentration–time data by finite differences:

initial point	$\frac{-3C_0 + 4C_1 - C_2}{2\Delta t}$
interior point	$\frac{C_{i-1} - C_{i+1}}{2\Delta t}$
final point	$\frac{C_{N-2} - 4C_{N-1} + 3C_N}{2\Delta t}$

5.2 Integral method

- **Guess the order, integrate, and test for a straight line.** The order that linearises the data is the order.

order 0	C_A against t is linear
order 1	$\ln C_A$ against t is linear
order 2	$1/C_A$ against t is linear
$A + B$, $M = C_{B0}/C_{A0} \neq 1$	$\ln \frac{C_A}{C_B} = \ln M + C_{A0}(M - 1)kt$

5.3 Half-life method

$$t_{1/2} = \frac{2^{n-1} - 1}{(n-1)k} C_{A0}^{1-n} \quad \text{order } n \neq 1$$

$$\ln t_{1/2} = (1-n) \ln C_{A0} + \text{const} \quad \text{slope} = 1-n \implies n = 1 - \text{slope}$$

$$n = 1: \quad t_{1/2} = \frac{\ln 2}{k} \quad \text{independent of } C_{A0}$$

A half-life that does not change with initial concentration is the signature of first-order kinetics — no regression needed.

5.4 Differential reactor and initial rates

$$-r'_A = \frac{F_{A0} - F_A}{W} \approx \frac{F_P}{W}$$

low conversion, so composition is nearly uniform

5.5 Nonlinear regression and enzyme kinetics

$$s^2 = \sum_i (C_{m,i} - C_{c,i})^2$$

minimise; m = measured, c = calculated

$$-r_S = \frac{k_3 C_S C_E}{C_S + C_M}$$

Michaelis–Menten

$$k_3 C_E t = (C_{S0} - C_S) + C_M \ln \frac{C_{S0}}{C_S}$$

integrated form

5.6 Variable-volume batch, constant pressure

$$X = \frac{(V/V_0) - 1}{\varepsilon_A}$$

read conversion straight off the volume change

5.7 Worked from the deck

Problem	Result
Trityl + methanol (regression of $\ln t_{1/2}$)	$n = 2.14$; $k' = 0.1243$; $k = 0.249 \text{ dm}^6 \text{ mol}^{-2} \text{ min}^{-1}$
Dimethyl ether, constant volume	$p_{i0} = 310.3$; $k = 4.4, 4.3, 4.4, 4.9 \times 10^{-4} \text{ s}^{-1}$, mean 4.50×10^{-4}
$A \rightleftharpoons R$, reversible	$k_1 = 0.05776$, $k_2 = 0.02888$
L5.19	$n = 2.00$; $k = 0.0040, 0.0040, 0.00402, 0.00401 \text{ L mmol}^{-1} \text{ min}^{-1}$
L3.25	$k = 0.1604 \text{ min}^{-1}$
Bomb reactor, $2A \rightarrow B$	1st order $k = 0.0517$, $r^2 = 0.886$; 2nd order $k = 0.0741$, $r^2 = 0.964$
Six MFRs in series vs one PFR	$X_{\text{MFRs}} = 0.750$, $X_{\text{PFR}} = 0.667$
Two-reactor split	$X_A = 0.2308$, $X_B = 0.1154$, $C_B = 46.00$
Sucrose inversion, Michaelis–Menten	$k_3 C_{S0} = 0.3210 \text{ h}^{-1}$; $C_M = 0.5151 \text{ mmol/L}$; $k_3 = 32.1 \text{ h}^{-1}$

- **The bomb-reactor comparison is the cleanest illustration of the integral method in the whole deck.** Both orders were fitted to the same data; the second-order fit has $r^2 = 0.964$ against 0.886 for first order. The higher r^2 is what selects the order — not a preference, and not the value of k .
- **Reversible PFR** (slide 61), $A \rightleftharpoons R$:

$$X_e = 0.80, \quad \tau = 20.0 \text{ min}, \quad C_{Af} = 0.0494 \quad \implies \quad \boxed{X = 0.506}$$

$X = 0.506$ against an equilibrium ceiling of $X_e = 0.80$: the reactor is approaching equilibrium, not running out of residence time. Adding volume beyond this point buys very little, which is the practical significance of computing X_e first.

6 Multiple Reactions

Source: Ch6_krd.pdf.

6.1 The four classes of multiple reaction

- **Parallel (competing):** $A \xrightarrow{k_1} B$ and $A \xrightarrow{k_2} C$ consume the same feedstock at once; the reactor's job is to steer consumption toward one branch.
- **Series (consecutive):** $A \xrightarrow{k_1} B \xrightarrow{k_2} C$ — the product of the first reaction is the feed of the second, so the intermediate B passes through a maximum.
- **Complex (parallel + series):** e.g. $A + B \xrightarrow{k_1} C + D$ together with $A + C \xrightarrow{k_2} E$; the deck also prints the **Denbigh network** $A \xrightarrow{1} R \xrightarrow{3} S$ with the side arrows $A \xrightarrow{2} T$ and $R \xrightarrow{4} U$ — the standard bookkeeping test for net rates.
- **Independent:** $A \rightarrow B + C$ and $D \rightarrow E + F$ run side by side without sharing species.

The deck's own chemistry: parallel — ethylene + oxygen burning to $2\text{CO}_2 + 2\text{H}_2\text{O}$ (k_1) versus epoxidation to ethylene oxide (k_2); series — ethylene oxide + NH_3 to mono-, then di-, then tri-ethanolamine; complex — ethanol to ethylene

and to acetaldehyde, then ethylene + acetaldehyde to butadiene + water; independent — cracking of crude, $C_{15}H_{32} \rightarrow C_{12}H_{26} + C_3H_6$ and $C_8H_{18} \rightarrow C_6H_{14} + C_2H_4$.

With two reactants the deck's first lever is the *contacting pattern* (slide 5):

- C_A, C_B both high — charge A and B all at once (batch);
- C_A, C_B both low — add both slowly to a well-mixed vessel (semibatch / fed CSTR);
- C_B high, C_A low — start with B and feed A slowly.

The economic incentive (slide 8): with D desired and U undesired, maximise D and minimise U ; the deck's cost sketch shows the reaction cost falling and the separation cost rising as the system is pushed harder, so the *total* cost has a minimum in between — selectivity, not conversion alone, sets the optimum.

6.2 Selectivity and yield — the four definitions

$S_{D/U} = \frac{r_D}{r_U}$	instantaneous (rate) selectivity
$\tilde{S}_{D/U} = \frac{F_D}{F_U}, \frac{N_D}{N_U}$ (batch)	overall selectivity
$Y_D = \frac{r_D}{-r_A}$	instantaneous yield
$\tilde{Y}_D = \frac{F_D}{F_{A0} - F_A}, \frac{N_D}{N_{A0} - N_A}$ (batch)	overall yield

- **In a CSTR the two selectivities coincide** (slide 11): $F_D = r_D V$ and $F_U = r_U V$, so $\tilde{S}_{D/U} = r_D V / r_U V = S_{D/U}$.
- **Definitions differ between authors** — slide 14 says to check which one a question intends. The *overall* quantities set the profit of the reaction–separation system; the *instantaneous* ones are the design tool, because they say which reactor and contacting scheme to choose.

6.3 The rate selectivity parameter, one reactant

For $A \xrightarrow{k_D} D$ (desired) and $A \xrightarrow{k_U} U$ (undesired) with $-r_A = k_D C_A^{\alpha_1} + k_U C_A^{\alpha_2}$:

$S_{D/U} = \frac{r_D}{r_U} = \frac{k_D}{k_U} C_A^{\alpha_1 - \alpha_2}$	maximise this
$\frac{k_D}{k_U} = \frac{A_D}{A_U} e^{-(E_D - E_U)/RT}$	the temperature lever

Case 1 $\alpha_1 > \alpha_2$ keep C_A **high**: batch or plug flow, no inerts and high pressure (gas), minimum diluent (liquid)

Case 2 $\alpha_2 > \alpha_1$ keep C_A **low**: CSTR or recycle reactor (the product stream acts as diluent), inerts and low pressure (gas), maximum diluent (liquid)

Case 3 $E_D > E_U$ run **hot**: k_D grows faster with T than k_U

Case 4 $E_U > E_D$ run **cool** — but not so cool that the desired reaction stops proceeding

The deck's rules of thumb (slide 4) are the same table in words: a *high* reactant concentration favours the reaction of *higher* order, a *low* concentration the reaction of *lower* order, and for reactions of the *same* order concentration has no effect on the distribution.

- **Worked selectivity argument** (slides 18–19). Three parallel paths from A :

$$r_D = 0.0002 e^{36\,000(\frac{1}{300} - \frac{1}{T})} C_A \quad r_U = 0.0018 e^{25\,000(\frac{1}{300} - \frac{1}{T})} C_A^{1.5} \quad r_Q = 0.00452 e^{5\,000(\frac{1}{300} - \frac{1}{T})} C_A^{0.5}$$

$S_{D/Q} = \frac{r_D}{r_Q} \approx \text{very large at high } T$	$(E_D = 36\,000 \gg E_Q = 5\,000)$
$S_{D/U} = \frac{r_D}{r_U} = \frac{0.11 e^{11\,000(\frac{1}{300} - \frac{1}{T})}}{C_A^{0.5}}$	$(\frac{0.0002}{0.0018} = 0.11, \quad 36\,000 - 25\,000 = 11\,000, \quad 1 - 1.5 = -\frac{1}{2})$

Two levers survive the algebra: **high temperature** (to bury Q) and **low concentration of A** (to bury U) — the latter by inerts, low pressure, or a CSTR / recycle reactor. One printed sentence runs against the algebra: slide 18's " S_{DU} is minimized at low concentration of A" states the reverse of the $C_A^{-0.5}$ dependence of the slide's own equation (5); slide 19's operating conclusions agree with the algebra, and the sentence is reported as printed.

6.4 Two reactants: signs, schemes and the four cases

For $A + B \xrightarrow{k_1} D$ and $A + B \xrightarrow{k_2} U$ with $r_D = k_1 C_A^{\alpha_1} C_B^{\beta_1}$, $r_U = k_2 C_A^{\alpha_2} C_B^{\beta_2}$:

$$S_{D/U} = \frac{k_1}{k_2} C_A^{\alpha_1 - \alpha_2} C_B^{\beta_1 - \beta_2}$$

The deck's menu of contacting schemes (slide 21): (a) CSTR; (b) tubular reactor; (c) batch; (d) semibatch, B fed slowly into A; (e) semibatch, A fed slowly into B; (f) tubular reactor with side streams of B; (g) tubular reactor with side streams of A; (h) series of small CSTRs; (i) tubular reactor with recycle.

Case I $\alpha_1 > \alpha_2$, $\beta_1 > \beta_2$	keep C_A, C_B both high — (b) tubular, (c) batch, high pressure, reduce inert
Case II $\alpha_1 > \alpha_2$, $\beta_1 < \beta_2$	C_A high, C_B low — (d) semibatch B into A, (f) side streams of B, (h) small CSTRs with A fed only to the first
Case III $\alpha_1 < \alpha_2$, $\beta_1 < \beta_2$	keep C_A, C_B both low — (a) CSTR, (i) large recycle ratio, feed diluted with inert, low pressure
Case IV $\alpha_1 < \alpha_2$, $\beta_1 > \beta_2$	C_A low, C_B high — (e) semibatch A into B, (g) side streams of A, (h) small CSTRs with fresh B to each

A slip in the source, reported rather than corrected. Slide 25 (Case IV) opens with the sentence "maintain the concentration of both A and B as high as possible," which contradicts its own $S_{D/U} = k_1 C_B^b / (k_2 C_A^a)$ with $a, b > 0$; the schemes it then lists — (e), (g), (h) with fresh B — are exactly the low- C_A /high- C_B set quoted in the table above.

6.5 Series reactions — maximising the intermediate

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C \quad \Rightarrow \quad C_B = C_{A0} \frac{k_1}{k_2 - k_1} (e^{-k_1 t} - e^{-k_2 t})$$

$$t_{\text{opt}} = \frac{\ln(k_2/k_1)}{k_2 - k_1} \quad \text{maximises } C_B$$

- In **parallel** networks the product is maximised by adjusting the reaction *conditions* and choosing the proper reactor; in **series** networks the lever is *time* — the space time of a flow reactor or the real time of a batch. The $C_B(t)$ form above is the one behind the Ch4 series numerical reproduced in §4.6 (Q20).
- The deck's triangular diagram (slide 27): $k_1/k_2 \gg 1$ gives a large quantity of B; $k_1/k_2 \sim 1$ a moderate one; $k_1/k_2 \ll 1$ (first reaction slow, second fast) very little. In every case a *long* reaction time in batch or a *long* tubular reactor converts B onward to C — which is why the optimum is interior.
- Verified numerically in §4.6: $t_{\text{max}} = 0.693$ h gives $C_{B,\text{max}} = 1.250$.

6.6 Product distribution — the quantitative treatment

$\varphi = \frac{dC_R}{-dC_A}$	instantaneous fractional yield of R
$\Phi = \frac{C_{Rf}}{C_{A0} - C_{Af}}$	overall yield = $\bar{\varphi}$ in the reactor
PFR: $\Phi_p = \frac{1}{\Delta C_A} \int_{C_{A0}}^{C_{Af}} \varphi dC_A$	area under $\varphi(C_A)$
MFR: $\Phi_m = \varphi$ evaluated at C_{Af}	one rectangle
N staged MFRs: $\Phi_N = \frac{\sum_i \varphi_i (C_{A,i-1} - C_{A,i})}{C_{A0} - C_{AN}}$	steps between the two
$C_{Rf} = \Phi (C_{A0} - C_{Af})$	product out of any reactor

Slide 31 draws the three constructions on one φ curve: plug flow shades the area under the curve from C_{Af} to C_{A0} , mixed flow draws the single rectangle of height $\varphi(C_{Af})$ over $(-\Delta C_A)$, and staged reactors climb the curve in steps — the rectangle construction converging to the area as $N \rightarrow \infty$.

6.7 Net rates and the bookkeeping for q reactions

$$r_j = \sum_{i=1}^q r_{ij} \quad \text{net rate of species } j = \text{sum over reactions } i$$

- The deck's example set (slide 34): reaction 1, $A + B \rightarrow 3C + D$; reaction 2, $A + 2C \rightarrow 3E$; reaction 3, $2B + 3E \rightarrow 4F$; reaction q , $A + \frac{1}{2}C \rightarrow G$. Then $r_A = r_{1A} + r_{2A} + r_{3A} + \dots + r_{qA}$ while r_B picks up only reactions 1 and 3 — species absent from a reaction contribute zero to its sum.

6.8 Worked from the deck

- Product distribution** (slide 32). $dC_R/dt = 1.0 C_A^{1.5} C_B^{0.3}$, $dC_S/dt = 1.0 C_A^{0.5} C_B^{1.8}$; equal volumetric feeds each at 20 mol/L, 90% conversion of A. The same chemistry through three contacting patterns:

$$(a) \text{ plug flow } C_R = 2.859 \quad (b) \text{ mixed flow } C_R = 4.50 \quad (c) \text{ plug flow, B in side streams } C_R = 7.295$$

With equal feeds $C_A = C_B$ everywhere, so $r_R/r_S = C^{-0.5}$: low concentration favours R. Mixed flow then runs entirely at the lowest concentration, plug flow must pass through the high-concentration region making S, and side entrances of B depress C_B relative to C_A further still — hence $2.86 < 4.50 < 7.30$. A factor of 2.6 in product from contacting alone, with no change to chemistry, temperature or throughput.

- Two MFRs in series** (slide 33), $r_R = k_1 C_A^2$, $r_S = k_2 C_A$, feed $C_{A0} = 1.0$, $C_{S0} = 0.3$, $\tau_1 = 2.5$, $\tau_2 = 10$ min, first-reactor exit $C_{A1} = 0.4$, $C_{R1} = 0.2$, $C_{S1} = 0.7$:

$$C_{A2} = 0.0745, \quad C_{R2} = 0.2277, \quad C_{S2} = 0.9978$$

7 Non-Isothermal Reactor Design

Source: Ch7-_krd.pdf.

7.1 Performance equations and the energy balance

- The isothermal algorithm of section 4 survives unchanged; temperature only adds one unknown, coupled in through Arrhenius. Slide 5 writes the set for a first-order $A \rightarrow B$ PFR:

$$\frac{dX}{dV} = \frac{-r_A}{F_{A0}}, \quad -r_A = k_1 \exp \left[\frac{E}{R} \left(\frac{1}{T_1} - \frac{1}{T} \right) \right] C_{A0}(1-X)$$

- The open-system energy balance (slides 6–8) reads *accumulation = heat added – work done + energy in by flow – energy out by flow*. With $E_i \approx U_i$ (potential and kinetic energies dropped) and flow work absorbed into the enthalpy, steady state gives

$$\dot{Q} - \dot{W}_s + \sum F_{i0} H_{i0} - \sum F_i H_i = 0$$

$$H_i(T) = H_i^\circ(T_R) + C_{p,i} (T - T_R) \quad \text{no phase change, constant } C_p$$

- Substituting the stoichiometry $F_i = F_{A0}(\Theta_i + \nu_i X/a)$ collapses everything to the deck's master equation (slides 9–11):

$$\dot{Q} - \dot{W}_s - F_{A0} X \left[\Delta H_{Rx}^\circ(T_R) + \Delta \hat{C}_p (T - T_R) \right] - F_{A0} \sum_i \Theta_i \tilde{C}_{p,i} (T - T_{i0}) = 0$$

$$\Delta \hat{C}_p = \frac{d}{a} C_{pD} + \frac{c}{a} C_{pC} - \frac{b}{a} C_{pB} - C_{pA}$$

7.2 Adiabatic operation

$$X = \frac{\sum \Theta_i \tilde{C}_{p,i} (T - T_0)}{-\left[\Delta H_{Rx}^\circ(T_R) + \Delta \hat{C}_p (T - T_R)\right]} \quad \text{the energy-balance line } X_{EB}$$

$$T = \frac{X [-\Delta H_{Rx}^\circ(T_R)] + \sum \Theta_i C_{p,i} T_{i0} + X \Delta C_p T_R}{\sum \Theta_i C_{p,i} + X \Delta C_p} \quad \text{solved for } T$$

- The same straight line X_{EB} on the T - X plane holds for **batch, CSTR, PFR and PBR** — the deck brackets all four beside the plot. For an exothermic reaction X_{EB} rises with T ; the reactor can only operate where this line meets what the kinetics and equilibrium allow.

7.3 The equilibrium ceiling, from thermodynamics

Slides 35–45 develop the ceiling rather than asserting it:

$$dG = V dp - S dT + \sum_j \mu_j dn_j, \quad \mu_j = \left(\frac{\partial G}{\partial n_j} \right)_{T,P,n_i}$$

$$\Delta G = \Delta G^\circ + RT \ln \left(\prod_j a_j^{\nu_j} \right), \quad K_a = \left(\prod_j a_j^{\nu_j} \right)_{eq} = \exp \left(\frac{-\Delta G^\circ}{RT} \right)$$

$$K_p = \left(\prod_j P_j^{\nu_j} \right), \quad K_f = \left(\prod_j f_j^{\nu_j} \right), \quad K_C = \left(\prod_j C_j^{\nu_j} \right) \quad \text{activity specialised}$$

$$\frac{d \ln K_a(T)}{dT} = \frac{\Delta H}{RT^2} \quad \text{van 't Hoff}$$

$$K_p = \left(\prod_j \left[P \frac{N_{j0} + \nu_j \alpha}{N_{T0} + \alpha \sum_j \nu_j} \right]^{\nu_j} \right) = F(\alpha) \quad \text{extent form}$$

- **Reversible exothermic:** X_e falls with T (the equilibrium curve descends); **reversible endothermic:** X_e rises with T . The adiabatic energy-balance line crosses the equilibrium curve; that intersection is the *adiabatic equilibrium conversion*, and no catalyst moves it (slide 46).
- **Worked intersection** (slides 47–48): $A \rightleftharpoons B$, pure A at 300 K, $H_{A0} = -40\,000$, $H_{B0} = -60\,000$ cal/mol, $C_{pA} = C_{pB} = 50$ cal/mol K, $K_C(298) = 10^5$:

$$X_{EB} = \frac{C_p(T - T_0)}{-\Delta H_{Rx}} = 2.5 \times 10^{-3}(T - 300) \quad \left(\frac{50}{20\,000} \right)$$

$$K_C(T) = 100\,000 \exp \left[-33.78 \frac{T - 298}{T} \right] \quad \left(\frac{|\Delta H|}{R \cdot 298} = \frac{10\,065}{298} = 33.78 \right)$$

$$X_e = \frac{K_C(T)}{1 + K_C(T)} \quad \text{intersection: deck graphic 0.42 at 465 K}$$

Solving the two printed equations exactly gives $X = 0.401$ at 460.4 K — the deck's 0.42/465 K is a graphical read-off of the same pair, reproduced by `krd-answers-check.py`. The deck's K_C table (slide 47: 661.6 at 350 K down to 0.12 at 500 K) is the same formula evaluated.

7.4 Heat exchange in a tubular reactor

$$\frac{dT}{dV} = \frac{U a (T_a - T) + (-r_A)(-\Delta H_{Rx})}{\sum F_i C_{p,i}} \quad \text{no radial gradients}$$

co-current coolant: $\frac{dT_a}{dV} = \frac{U a (T - T_a)}{\dot{m}_c C_{p,c}}$

- If the coolant flow is large compared with the heat released, T_a is effectively **constant** along the reactor (slide 26); heat added for endothermic, removed for exothermic.
- **Acetic anhydride** (slides 27–30): acetone cracking $\text{CH}_3\text{COCH}_3 \rightarrow \text{CH}_2\text{CO} + \text{CH}_4$, first order with $\ln k = 34.34 - 34\,222/T$, 7850 kg/h through a bank of 1000 1-inch schedule-40 tubes, inlet 1035 K, 162 kPa. Three cases: adiabatic — the endothermic reaction cools itself and *dies after* $\approx 3.5 \text{ m}^3$; constant $T_a = 1150$ K with $U = 110 \text{ J/m}^2\text{s K}$ — T dips then recovers and X climbs; variable heating medium — T_a slides down with T .

- **Ethane cracking at equilibrium** (slide 31): ethane + steam, 2.99 atm, 880 °C, 95 m tubes, 60 % conversion — the deck's specification sheet for the equilibrium-limited endothermic case.

7.5 The CSTR with heat exchange — two solution routes

With $\kappa = UA/(F_{A0}C_{p0})$ and $T_c = (\kappa T_a + T_0)/(1 + \kappa)$:

Specify X $T = T_c + \frac{(-\Delta H_{Rx}^\circ)(X)}{C_{p0}(1 + \kappa)}$, then $k = Ae^{-E/RT}$, $-r_A = kC_{A0}(1 - X)$, $V = F_{A0}X/(-r_A)$

Specify T $X = \frac{C_{p0}(1 + \kappa)(T - T_c)}{-\Delta H_{Rx}^\circ}$, then the same rate-law chain to V

Specify V generate T vs X_{EB} from the energy balance and T vs X_{MB} from $X_{MB} = \tau k/(1 + \tau k)$; the intersection is the operating point

$$G(T) = X(-\Delta H_{Rx}) = \frac{\tau A e^{-E/RT}}{1 + \tau A e^{-E/RT}} (-\Delta H_{Rx}) \quad \text{heat generated, first order}$$

$$R(T) = C_{p0}(1 + \kappa)(T - T_c) \quad \text{heat removed}$$

- Raising T_0 slides the straight $R(T)$ line to the right; raising κ steepens it and moves T_c from T_0 ($\kappa = 0$) toward T_a ($\kappa \rightarrow \infty$) (slides 77–78). Raising τ or E sharpens the S-shape of $G(T)$ (slide 81) — which is what creates multiple intersections.

7.6 Multiple steady states, ignition–extinction, runaway

- Intersections of $G(T)$ and $R(T)$ are the steady states; sweeping T_0 traces the **ignition–extinction curve**: upper and lower branches are attainable, the middle branch is not (slides 82–84).
- **Stability** (slides 85–89): a state is stable where a temperature bump removes more heat than it generates. The deck's criterion: with

$$\Delta T_{bc} = T' - T_c = \frac{RT_c^2}{E}, \quad T' = \frac{E}{2R} \left[1 - \sqrt{1 - 4 \frac{RT_c}{E}} \right], \quad T_c = \frac{\kappa T_a + T_0}{1 + \kappa},$$

exceeding ΔT_{bc} tips the reactor to the upper state. For industrial reactions $E/RT \approx 16$ –24, so $\Delta T_{bc} \approx 15$ –30 °C.

- With $S' = dG/dT$ at the steady state, **runaway** when $C_{p0}(1 + \kappa) < S'$ (slides 88, 90–91): decreasing UA flattens $R(T)$ until tangency disappears and the CSTR runs away to the upper state. The deck demonstrates it on $2\text{Na}_2\text{S}_2\text{O}_3 + 4\text{H}_2\text{O}_2 \rightarrow \text{Na}_2\text{S}_3\text{O}_6 + \text{Na}_2\text{SO}_4 + 4\text{H}_2\text{O}$ (slides 85–86).
- **Zero vs first order** (slides 92–93): with $\tau = 0.1$ min, $E = 10\,000$ cal/mol, $\Delta H_{Rx} = -10\,000$ cal/mol and inerts 2:1, the deck's table gives $dE/dT = 80$; the first-order reaction is globally stable for all T_c , while the zero-order one runs away above an entering temperature of 460 K and becomes stable again only past 685 K.

7.7 Interstage cooling, interstage heating, optimum feed temperature

- **Exothermic reversible**: several adiabatic stages with *coolers* between them; each stage starts cooler, further below its equilibrium curve, and the zigzag path on the X – T diagram climbs above what one adiabatic bed can reach (slides 49, 52).
- **Endothermic reversible**: mirror image — *heaters* between stages (slide 50).
- **Optimum feed temperature** (slides 53–54): at low T_0 the rate is slow and conversion is low; at high T_0 the equilibrium conversion collapses and conversion is again low; between them $X(T_0)$ passes through a *maximum* — the optimum feed temperature for a fixed size or catalyst weight.

7.8 Multiple reactions with heat effects (PFR/PBR)

$$\frac{dT}{dV} = \frac{Ua(T_a - T) + \sum_i (-r_{ij})(-\Delta H_{Rx,ij})}{\sum_j F_j C_{p,j}} \quad \text{coupled with the mole balances and rate laws}$$

Slide 95's example: $A \rightarrow B$ ($-r_{1A} = k_{1A}C_A$) and $2A \rightarrow C$ ($-r_{2A} = k_{2A}C_A^2$), pure A at 100 mol/s, 150 °C, 0.1 mol/dm³; $\Delta H_1 = -20\,000$, $\Delta H_2 = -60\,000$ J/mol A; $C_{pA} = C_{pB} = 90$, $C_{pC} = 180$ J/mol °C; $k_{1A} = 10 \exp[(E_1/R)(1/300 - 1/T)]$ s⁻¹ with $E_1/R = 4000$ K; $k_{2A} = 0.09 \exp[(E_2/R)(1/300 - 1/T)]$ dm³/mols with $E_2/R = 9000$ K; $Ua = 4000$ J/m³s °C, $T_a = 100$ °C.

7.9 Worked from the deck

- **Butane isomerisation, adiabatic PFR** (slides 14–24). 163 kmol/h of 90 % n-butane/10 % i-pentane at 330 K to 70 %; $\Delta H_{Rx} = -6900$ J/mol, $E/R \approx 7906$ K, $C_{p,\text{butane}} = 141$, $C_{p,\text{pentane}} = 161$ J/mol K. The energy balance gives $T = 330 + 43.3X$ (slope $6900/158.9 \approx 43.3$), and the deck's own explicit set is

$$K_C = 3.03 e^{-830.3(T-333)/(T-333)}, \quad k = 31.1 e^{7906(T-360)/(T-360)}$$

$$X_e = \frac{K_C}{1 + K_C}, \quad -r_A = kC_{A0} \left(1 - \left(1 + \frac{1}{K_C}\right)X\right)$$

X	0	0.2	0.4	0.6	0.65	0.7
T (K)	330	338.7	347.3	356.0	358.1	360.3
k (h ⁻¹)	4.22	7.76	14.02	24.27	27.74	31.67
X_e	0.76	0.74	0.73	0.72	0.718	0.715
$F_{A0}/(-r_A)$ (m ³)	3.74	2.78	2.50	3.88	5.99	23.29

$$V_{X=0.70} \approx 1.97 \text{ m}^3$$

the rate peaks mid-reactor ($-r_A \approx 59$ at $X \approx 0.4$) then collapses toward X_e

- **PFR with co-current heat exchange** (slides 61–67): same chemistry with $Ua = 5000$ kJ/h m³K, inlet 305 K, coolant 310 K; the deck integrates $dT_a/dV = 5000(T - T_a)/14\,000$ alongside $dT/dV = (r_A\Delta H - Ua(T - T_a))/(C_{p0}F_{A0})$ — the rate peak is cut and T never exceeds the 325 K limit.
- **Interstage cooling** (slides 48–52), 40 mol/s cooled to 350 K between stages, each stage to 95 % of its equilibrium conversion:

$$\text{R1: } X = 0.397 \text{ at } 458.9 \text{ K; } \quad \text{R2: } X = 0.621 \text{ at } 439.5 \text{ K}$$

$$\dot{Q}_1 = 217.8 \text{ kcal/s, } \dot{Q}_2 = 179.1 \text{ kcal/s}$$

The second stage exits *cooler* than the first (439.5 vs 458.9 K) yet reaches *higher* conversion — the direct payoff of resetting the distance to equilibrium.

- **Propylene oxide, CSTR** (slides 57–74). 43.04 lbmol/h PO into the 300-gal CSTR, first order with

$$k = 16.96 \times 10^{12} e^{-32\,400/(1.987 T)} \text{ h}^{-1}, \quad \tau = 0.1229 \text{ h,}$$

plant ceiling 125 °F:

Case	X	T (R)	T (F)
adiabatic	0.842	610.8	151
with cooling coil	0.364	563.7	104

The adiabatic point sits above the 125 °F = 585 R ceiling — “we can’t use the existing system.” The 40 ft² coil at 85 °F with $U = 100$ Btu/h ft² F pulls the operating point down to the deck’s fsolve values $X = 0.364$, $T = 563.7$ R, inside the constraint. The X_{MB}/X_{EB} table of slide 59 crosses at 613 R, 0.85 — the graphical version of the adiabatic row.

- **Parallel reactions with heat effects** (slide 95). The coupled ODE set of §7.8 integrated to $V = 10^4$ dm³ gives $T \rightarrow 645$ K with the product essentially B.

8 Catalysis and Catalytic Reactors

Source: Catalysis_Krd.pdf.

8.1 The seven steps and the rate-limiting-step algorithm

External diffusion → internal diffusion → adsorption of A → surface reaction → desorption of B → internal diffusion out → external diffusion out. Assume one step limits, write its rate, and combine with the site balance

$$C_t = C_v + \sum_i C_{i,S}$$

Then test the assumption: if the assumed rate-limiting step does not reproduce the data, move to the next one. The deck works through each assumption in turn.

8.2 Adsorption isotherms and rate forms

$$C_{A,S} = \frac{K_A P_A C_t}{1 + K_A P_A + K_B P_B} \quad \text{Langmuir, competitive sites}$$

$$-r'_T = \frac{k P_T P_{H_2}}{1 + K_B P_B + K_T P_T} \quad \text{Eley-Rideal type}$$

- **Langmuir-Hinshelwood** assumes both reactants are adsorbed before reacting; **Eley-Rideal** assumes one reacts directly from the gas phase. The denominator is the site balance in both cases — that is what makes the rate saturate at high pressure.

8.3 Internal diffusion — Thiele modulus and effectiveness factor

$$\phi_s = R \sqrt{\frac{k_1}{D_e}} \quad \text{Thiele modulus, first order, sphere}$$

$$\eta = \frac{3}{\phi^2} (\phi \coth \phi - 1) \quad \text{effectiveness factor, sphere}$$

$$\eta = \frac{\tanh \phi}{\phi} \quad \text{slab geometry}$$

$$\frac{C_A}{C_{As}} = \frac{R}{r} \cdot \frac{\sinh(\phi r/R)}{\sinh \phi} \quad \text{concentration profile inside the pellet}$$

- $\eta \rightarrow 1$ for small ϕ (reaction-limited: the whole pellet is used); $\eta \rightarrow 3/\phi$ for large ϕ (diffusion-limited: only a shell near the surface reacts).

8.4 Diagnostic criteria

$$C_{WP} = \eta \phi^2 = \frac{(-r'_A)_{\text{obs}} R^2}{D_e C_{As}} \quad \text{Weisz-Prater}$$

$$C_{WP} \ll 1 \quad \text{no internal limitation}$$

$$C_{WP} \gg 1 \quad \text{strongly limited}$$

$$\Omega = \frac{\eta}{1 + \eta \phi^2 S_a \rho_b / k_c} \quad \text{overall effectiveness, with external resistance}$$

$$\frac{k C_{Ab}^{n-1} R^2 \rho_b S_a E}{b D_e T^2} < 0.15 \quad \text{Mears: external gradients negligible}$$

Weisz-Prater is the practical one: it uses the *observed* rate, so it can be evaluated from experiment without knowing the intrinsic kinetics.

8.5 Worked from the deck

- **Concentration inside a pellet** (slide 48), sphere $d_p = 2 \times 10^{-3}$ cm, $D_e = 0.1$ cm²/s:

$$\phi = 5.986, \quad \frac{C_A}{C_{As}} = 0.2371, \quad C_A = 2.37 \times 10^{-4}$$

$$\text{to reach } C_A/C_{As} = 0.8 : \quad \phi = 2.042 \quad \implies \quad \boxed{d_{\text{new}} = 6.82 \times 10^{-4} \text{ cm}}$$

At $\phi = 5.99$ only 24 % of the surface concentration survives at the pellet centre. Shrinking the diameter to about a third of the original is what it takes to make the whole pellet effective.

- **Apparent activation energy** (slide 57), $R = 0.2 \text{ cm}$, $D_e = 0.015 \text{ cm}^2/\text{s}$, $100 \rightarrow 150 \text{ }^\circ\text{C}$:

T	ϕ	η	r
100 °C	1.575	0.8660	2.617×10^{-5}
150 °C	7.75	0.337	2.468×10^{-4}

$$\boxed{E_{\text{app}} = 14\,080 \text{ cal/mol}}$$

The measured activation energy is roughly *half* the true one because η is falling as fast as k is rising — the classic signature of internal diffusion control. Halving the observed E is the diagnostic, not an error in the Arrhenius plot.

- **Reactor choice:** the packed bed is the workhorse; the deck covers adiabatic and non-adiabatic operation, reverse-flow and counter-flow configurations.

9 Non-Ideal Reactors and Residence-Time Distributions

Source: Non_ideal_KRD.pdf — headed *Reaction Engineering / CHE G641*, the graduate treatment of this material.

9.1 Where ideality breaks

- **Why it matters (slides 3–4):** the ideal design equations assume identical residence times (PFR/PBR, no axial diffusion) or perfect mixing (CSTR/batch); real vessels sit in between — non-uniform flow, inefficient or absent mixing, bypassing and dead zones change conversion, yield and the temperature development of the process.
- **CSTR (slide 5):** the feed entrance runs at relatively high concentration while stagnant *dead zones* run low; part of the flow can short-circuit the vessel as bypass.
- **PBR (slide 6):** ideal plug flow moves every molecule at the same axial rate; the real bed adds turbulent velocity profiles, back-mixing and molecular diffusion, so molecules travel at different speeds and in different directions (channeling, slide 14).
- **Micro vs macro mixing (slides 8–9):** macromixing is the RTD itself; micromixing asks whether molecules of different ages meet at all — *complete segregation*, same-age globules leaving together, versus *complete micromixing* — and whether mixing occurs early or late along the flow path.

9.2 The RTD functions

$E(t) = \frac{C(t)}{\int_0^\infty C \, dt}$	normalised tracer response
$F(t) = \int_0^t E \, dt$	cumulative fraction younger than t
$t_m = \int_0^\infty t E \, dt = \frac{V}{v_0}$	mean residence time
$\sigma^2 = \int_0^\infty (t - t_m)^2 E \, dt$	variance, i.e. the spread
$1 - F(t) = \frac{V}{v} I(t)$	internal age distribution $I(t)$

Normalising by the area is what makes $E(t)$ a probability density, and it is the step most often missed when a pulse experiment is worked up by hand.

The profiles are a diagnostic (slides 14, 21): a sharp spike reads as a nearly ideal PFR, a decaying exponential as a nearly ideal CSTR, an early spike with a long tail as a PBR with channeling and dead zones, and a two-humped curve as a CSTR with dead zones.

9.3 Ideal limits

PFR: $E(t) = \delta(t - \tau)$	all fluid elements the same age
CSTR: $E(t) = \frac{e^{-t/\tau}}{\tau}$	exponential; some fluid leaves almost at once

9.4 Zero-parameter models: segregation and maximum mixedness

$$\bar{X} = \int_0^\infty X_{\text{batch}}(t) E(t) dt \quad \text{each fluid element is a small batch reactor}$$

- $n > 1$: $\bar{X}$ exceeds the ideal-mixed conversion. $0 < n < 1$: the opposite. Complete segregation is therefore a bound, and it tells you which way a real reactor departs from ideal mixing without any further modelling.

The deck's model menu (slide 22): *zero* adjustable parameters — the segregation model above and the maximum-mixedness model, its micromixed opposite extreme; *one* parameter — tanks-in-series and dispersion; *two* parameters — the real vessel as a combination of ideal reactors (§9.7).

9.5 Tanks-in-series model

$$E(t) = \frac{t^{n-1}}{(n-1)! \tau_i^n} e^{-t/\tau_i} \quad \text{gamma distribution}$$

$$n = \frac{t_m^2}{\sigma^2} \quad \text{number of tanks, from the tracer data alone}$$

$$X = 1 - (1 + k\tau_i)^{-n} \quad \text{first-order conversion}$$

Where the gamma form comes from (slides 26–28): a pulse through N equal CSTRs gives $C_1 = C_0 e^{-t/\tau_i}$, $C_2 = C_0 t e^{-t/\tau_i} / \tau_i$, $C_3 = C_0 t^2 e^{-t/\tau_i} / 2\tau_i^2$, and in general $E(t) = t^{N-1} e^{-t/\tau_i} / [(N-1)! \tau_i^N]$, or dimensionless $E(\theta) = N(N\theta)^{N-1} e^{-N\theta} / (N-1)!$. As $N \rightarrow \infty$ the curve collapses to the plug-flow spike, which is why $n = t_m^2 / \sigma^2$ measures the departure from plug flow.

9.6 Dispersion model

$$X = 1 - \frac{4q e^{Pe/2}}{(1+q)^2 e^{Pe/2} - (1-q)^2 e^{-Pe/2}} \quad \text{closed vessel, first order}$$

$$q = \sqrt{1 + \frac{4Da}{Pe}}, \quad Da = k\tau, \quad Pe = \frac{UL}{Da}$$

$$\sigma_\theta^2 = \frac{2}{Pe} - \frac{2}{Pe^2} (1 - e^{-Pe}) \quad \text{link from tracer data to } Pe$$

$$n = \frac{Pe}{2} + 1 \quad \text{equivalent number of tanks}$$

The picture behind it (slides 29–33): molecules diffuse ahead of and lag behind the molar average velocity, i.e. a Fickian dispersion $D_a dC/dz$ superimposed on the bulk flow, so the tracer pulse broadens as it travels; the mole balance over a slice ΔZ with $F_A = uAC_A + AD_a dC_A/dz$ yields the closed-vessel solution above. The single parameter $D_a/(UL)$ is read from a pulse tracer test; the deck's E -curves for $D_a/(UL) = 0.002, 0.025, 0.2$ show small, intermediate and large deviation from plug flow.

9.7 Two-parameter models

- A real CSTR is represented as an **ideal reacting region** of volume $V_s = \alpha V$ with a **dead zone** $V_d = (1 - \alpha)V$, plus a **bypass** stream; $\beta = v_s/v_0$ is the fraction of the feed entering the reacting region (slides 40–41). The modelled conversion is

$$1 - X = \beta + \frac{(1 - \beta)^2}{1 - \beta + \alpha\tau k}.$$

- Tracer response takes the form

$$\frac{C_T}{C_{T0}} = 1 - (1 - \beta) e^{-(1-\beta)t/(\alpha\tau)}$$

and linearising it (slide 42),

$$\ln \frac{C_{T0}}{C_{T0} - C_T} = \ln \frac{1}{1 - \beta} + \frac{1 - \beta}{\alpha} \cdot \frac{t}{\tau},$$

so the intercept and slope of the tracer straight line give (α, β) .

- The deck's second two-parameter variant (slides 43–44) models a real CSTR as **two ideal CSTRs with interchange** — a highly agitated region near the impeller exchanging with a quieter one, $V_d = (1 - \alpha)V$ — with

$$X = \frac{(\beta + \alpha\tau k)(\beta + (1 - \alpha)\tau k) - \beta^2}{(1 + \beta + \alpha\tau k)(\beta + (1 - \alpha)\tau k) - \beta^2},$$

and (α, β) again fixed by a tracer test, the response now a sum of two exponentials with exponents m_1, m_2 .

9.8 Worked from the deck

- **Pulse tracer** (slide 17):

$$\text{area} = 50.03, \quad t_m = 5.19 \text{ min}, \quad \sigma^2 = 6.03 \quad \implies \quad \boxed{\text{fraction between 3 and 6 min} = 0.513}$$

Just over half the fluid leaves within a minute either side of the mean — the spread is the whole story here, not the mean.

- **Dispersion example** (slides 52–59):

$$Pe = 7.74, \quad Da = 1.28$$

Model	X
dispersion (closed vessel)	0.679
tanks-in-series, $n = 4.45$	0.676
ideal PFR	0.720
ideal CSTR	0.560

The two real-reactor models agree to 0.003 and sit *between* the ideal limits, as they must. Using $n = Pe/2 + 1 = 4.87$ instead of the moment-derived 4.45 changes the answer negligibly — but the agreement of the two independent models is what justifies trusting either. Treating this vessel as an ideal CSTR would understate conversion by 0.12; as an ideal PFR, overstate it by 0.04.

Formula Sheet — the whole course on one page

$$X = \frac{N_{A0} - N_A}{N_{A0}}$$

$$V_{\text{CSTR}} = \frac{F_{A0}X}{(-r_A)_{\text{exit}}}$$

$$W_{\text{PBR}} = F_{A0} \int_0^X \frac{dX}{-r'_A}$$

$$-r_A = kC_A^\alpha C_B^\beta$$

$$C_i = C_{A0} \frac{\Theta_i + \nu_i X/a}{1 + \varepsilon X} \cdot \frac{T_0}{T} \cdot \frac{P}{P_0}$$

$$X_{\text{CSTR, 1st}} = \frac{k\tau}{1 + k\tau}$$

$$t_{1/2} = \frac{2^{n-1} - 1}{(n-1)k} C_{A0}^{1-n}$$

$$S_{D/U} = \frac{r_D}{r_U}$$

$$T = T_0 + (\text{slope})X$$

$$\phi = R\sqrt{k_1/D_e}$$

$$C_{WP} = \frac{(-r'_A)_{\text{obs}} R^2}{D_e C_{As}}$$

$$\tau = \frac{V}{v_0}$$

$$V_{\text{PFR}} = F_{A0} \int_0^X \frac{dX}{-r_A}$$

$$t_{\text{batch}} = N_{A0} \int_0^X \frac{dX}{-r_A V}$$

$$k = Ae^{-E/RT}$$

$$\varepsilon = y_{A0}\delta$$

$$X_{N \text{ CSTRs}} = 1 - \frac{1}{(1 + k\tau_i)^N}$$

$$\ln t_{1/2} = (1 - n) \ln C_{A0} + \text{const}$$

$$\tilde{Y}_D = \frac{F_D}{F_{A0} - F_A}$$

$$X_e = \frac{K_C}{1 + K_C}$$

$$\eta = \frac{3}{\phi^2} (\phi \coth \phi - 1)$$

$$n = \frac{t_m^2}{\sigma^2} = \frac{Pe}{2} + 1$$

The four comparisons worth knowing cold

Question	Better	From
Same conversion, one vessel: PFR or CSTR?	PFR	§2.6 (2.165 vs 6.40 m ³)
Same total volume, two CSTRs in series or parallel?	series	§4.6 (0.900 vs 0.813)
PBR then CSTR, or CSTR then PBR?	CSTR first	§4.6 (0.690 vs 0.598)
Real vessel, or an ideal limit?	neither	§9.8 (0.679 vs 0.720 / 0.560)

Prepared from the nine CHE F311 slide decks named in the box on page 1; every formula above is the form taught in the deck heading its section, and every number is reproduced by `krd-answers-check.py`, which exits 0. Structure and depth follow the course standard set by `SP2-notes.tex` (CHE F313). Five places where the source itself is inconsistent are reported rather than corrected: the pressure-drop length in §4.6, the graduate (CHE G641) heading on `Non_ideal_KRD.pdf`, two Ch6 sentences — the concentration verb on slide 18 (§6.3) and the Case-IV opening on slide 25 (§6.4) — and the temperature-time pairing on Ch3 slide 61 (§3.5). Symbols: X conversion, $-r_A$ rate of disappearance of A, $-r'_A$ rate per catalyst mass, F_{A0} inlet molar flow of A, C_A concentration, Θ_i feed ratio C_{i0}/C_{A0} , ε volume-change factor, δ mole-change number, V volume, W catalyst mass, τ space time, Da Damköhler number, k rate constant, A pre-exponential factor, E activation energy, K_C equilibrium constant, X_e equilibrium conversion, ΔH_{Rx} heat of reaction, Ua heat-transfer coefficient times area, T_a coolant temperature, p reduced pressure P/P_0 , α pressure-drop parameter, ϕ bed porosity or Thiele modulus as the section states, η effectiveness factor, D_e effective diffusivity, C_{WP} Weisz-Prater number, $E(t)$ residence-time distribution, $F(t)$ cumulative RTD, t_m mean residence time, σ^2 variance, Pe Péclet number, n tanks-in-series number, $S_{D/U}$ selectivity, $\tilde{Y}_D$ yield.